ClaimGuard TPA — Technical Design Document (TDD)

Hospital: Company A Multispeciality Hospital, Nashik
Domain: Cashless Hospital Insurance Desk & TPA Operations
Regulatory Frameworks: IRDAI Master Circular (2024), DPDP Act (2023)
GitHub Repository: https://github.com/Prakhar6947/ClaimGuard
CI/CD Status: [![ClaimGuard CI](https://github.com/Prakhar6947/ClaimGuard/actions/workflows/ci.yml/badge.svg)](https://github.com/Prakhar6947/ClaimGuard/actions)

1. Problem Statement & Hospital Context

Company A Multispeciality Hospital in Nashik suffers significant cashless insurance revenue leakage (~■63.57 Lakhs annually) driven by:

1. **Pre-authorisation denials & deductions** caused by missing insurer-specific diagnostic evidence (e.g. Ultrasound report missing for gall bladder removal).

2. **IRDAI SLA breaches**: IRDAI mandates a strict 1-hour turnaround for cashless pre-authorisation and a 3-hour turnaround for final discharge bill approval. Inefficient communication cycles cause patient discharge delays and financial penalties.

3. **Manual document sorting overhead & transcription errors**: Desk executives manually read handwritten case papers and laboratory reports, introducing transcription errors and misfiled documents.

2. System Architecture

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- **Tier 3 (Insurer × Procedure Rules)**: Apply to specific procedure under specific insurer (e.g. Company B + Gall Bladder Lap Chole requires Abdominal Ultrasound Report; Company B + Total Knee Replacement requires Standing Knee X-Ray).

When a new admission is registered via `POST /api/v1/claims`, `sync_claim_checklist` automatically materializes these checklist items onto the claim.

4. Human-In-The-Loop AI Guardrails & DPDP Compliance

1. **Confidence Threshold (0.75)**: Any scan with classification confidence < 0.75 is quarantined into the Manual Review Tray.

2. **Pending Field Lock**: AI extracted values remain `PENDING` until a human Desk Executive clicks `Confirm` or edits the value. The Submission Gatekeeper prevents transmission while unconfirmed fields exist.

3. **Desk Head Waiver**: Mandatory slots can only be waived by users with the `DESK_HEAD` role, requiring a mandatory written justification recorded in the immutable audit log.

4. **DPDP PII Protection**: Policy numbers are masked by default (`XXXX-XXXX-last4`). Clicking "Reveal" triggers `POST /api/v1/claims/{id}/audit-unmask` and records an immutable audit log entry.

5. **ReportLab Consolidated PDF Packet**: Compiles all verified documents, checklist status, and patient demographics into a single merged packet matching insurer requirements.